

EE587 Nonlinear Control System

Class 1

Nonlinear Dynamical Systems

Ex. $F(x) = 0, x \in \mathbf{R}^n$

$$F(x) = Ax - b$$

F is affine in x

EE510: conditions on A and b

for existence and uniqueness of solution to $Ax = b$

$$A \in \mathbf{R}^{n \times n}, b \in \mathbf{R}^{n \times 1}$$

$$F(x) = Ax: \text{linear}$$

$$F(x) = Ax + b: \text{affine}$$

$$F(x) = x^T Ax: (\text{e.g., nonlinear})$$

Unconstrained optimization

$\min_x g(x)$ where $g: \mathbf{R}^n \rightarrow \mathbf{R}$

Necessary: $\nabla g(x) = 0 \rightarrow F(x) = 0$

$$g = \frac{1}{2} x^T Ax - bx$$

$$\nabla g = Ax - b$$

F nonlinear: in general difficult to solve $F(x) = 0$ analytically

Alternative: iterative approach

Gradient Descent

$$x^{k+1} = x^k - \alpha^k \nabla g(x^k) \text{ --- Discrete-time (DT)}$$

where α^k is step-size

$$\frac{x^{k+1} - x^k}{\alpha^k} = -\nabla g(x^k)$$

$$\frac{dx}{dt} = -\nabla g(x) \text{ --- Continuous-time (CT)}$$

$$\frac{dx}{dt} = -F(x)$$

$$\dot{x} = -F(x)$$

$$\implies \frac{x^{k+1} - x^k}{\alpha} = -F(x^k)$$

$$x^{k+1} = x^k - \alpha F(x^k)$$

$$x^{k+1} = -\tilde{F}(x^k)$$

where $\alpha F(x^k) - x^k = \tilde{F}(x^k)$

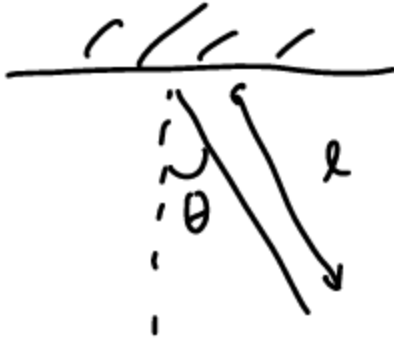
meaning $x^{k+1} \neq -\tilde{F}(x^k)$

$\frac{dx}{dt} = f(x)$ where $x \in \mathbf{R}^n, t: \text{time}, F: \mathbf{R}^n \rightarrow n$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \frac{dx}{dt} = \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \vdots \\ \dot{x}_n \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dots \\ \dot{x}_n \end{bmatrix} = f \left(\begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} \right) = \begin{bmatrix} f_1(x_1 \dots x_n) \\ \dots \\ f_n(x_1 \dots x_n) \end{bmatrix} \neq \begin{bmatrix} f_1(x_1) \\ \dots \\ f_n(x_n) \end{bmatrix}$$

Ex.



m : mass

F : external force

$$1. = 0$$

$$m \frac{d^2}{dt^2} + k \frac{d}{dt} + mg = 0$$

$$= x_1$$

$$\cdot = x_2$$

So,

$$\cdot = \dot{x}_1 = x_2$$

$$= \dot{x}_2$$

$$m\dot{x}_2 + kx_2 + mg x_1 = 0$$

$$\dot{x}_2 = -\frac{k}{m}x_2 - \frac{g}{m}x_1$$

$$\dot{x}_1 = x_2$$

$$= x_1$$

$$= x_2$$

$$f_1(x_1, x_2) = x_2$$

$$f_2(x_1, x_2) = -\frac{k}{m}x_2 - \frac{g}{m}x_1$$

$$= \begin{bmatrix} x_1 \\ x_2 \\ x \end{bmatrix}$$

$$x = t$$

$$\dot{x} = 1$$

$$f_1(x_1, x_2, x) = x_2$$

$$f_2(x_1, x_2, x) = -\frac{k}{m}x_2 - \frac{g}{m}x_1$$

$$f(x_1, x_2, x) = 1$$

$$2. = mg$$

$$m \frac{d^2}{dt^2} + k \frac{d}{dt} + mg = mg$$

$$m \frac{d^2}{dt^2} + k \frac{d}{dt} + mg(-) = 0$$

$$\dot{x} = f(x)$$

$$\begin{aligned} &= x_1 \\ &= x_2 \end{aligned}$$

$$x_1 =$$

$$x_2 = \cdot$$

$$f_1(x_1, x_2) = x_2$$

$$f_2(x_1, x_2) = -\frac{k}{m}x_2 - \frac{g}{m}x_1 + \frac{g}{m}x_1$$

$$\dot{x} = f(x): \text{no explicit}$$

$$\implies \text{autonomous system}$$

Forced system can still be autonomous!

$$3. = mg \, t$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{k}{m}x_2 - \frac{g}{m}x_1 + \frac{g}{m}t$$

$$\dot{x} = f(x) \text{ --- time-invariant}$$

$$\dot{x} = f(x, t) \text{ --- time-varying}$$

$$\dot{x} = f(x,)$$

Define:

$$x = t$$

$$\dot{x} = 1$$

$$m \frac{d^2}{dt^2} + k \frac{d}{dt} + mg =$$

$$= x_1$$

$$\cdot = x_2$$

$$\dot{x}_1 = x_2 = f_1(x_1, x_2,)$$

$$\dot{x}_2 = -\frac{k}{m}x_2 - \frac{g}{m}x_1 + \frac{g}{m}t = f_2(x_1, x_2,)$$

Specifying initial conditions - linear system

$$f: \mathbf{R}^n \times \mathbf{R}^m \rightarrow \mathbf{R}^n$$

where $x \in \mathbf{R}^n, \in \mathbf{R}^m$

$$\dot{x} = f(x)$$

$$x(0) = x_0$$

$$\dot{x} = Ax$$

$$f: \mathbf{R}^n \rightarrow \mathbf{R}^n, x \in \mathbf{R}^n, A \in \mathbf{R}^{n \times n}$$

$$\dot{x} = f(x,)$$

$$x(0) = x_0$$

$$\dot{x} = Ax +$$

$$x(0) = x_0$$

$$f(x,) = Ax +$$

$$f: \mathbf{R}^n \times \mathbf{R}^m \rightarrow \mathbf{R}^n, \in \mathbf{R}^m, \in \mathbf{R}^{n \times m}$$

$$x(t) = e^{At}x(0) + \int_{t_0}^{t} e^{A(t-\tau)}d\tau$$

where $e^{At}x(0)$ is the I.C., $\int_{t_0}^{t} e^{A(t-\tau)}d\tau$ is the input

$f(x)$ linear if:

$$-f(0) = 0$$

$$-f(\alpha x) = \alpha f(x)$$

$$-f(x_1 + x_2) = f(x_1) + f(x_2)$$

$$\dot{x} = -\alpha \frac{x^2}{k} = f(x)$$

$$x^2 =$$

$$f() = -\alpha \frac{x}{k}$$

Differentiate: $2x\dot{x} = \cdot$

$$\dot{x} = \frac{\cdot}{2x}$$

$$\frac{\cdot}{2x} = -\alpha \frac{x}{k}$$

$$\dot{x} = -\alpha \frac{x}{k}$$

$$\cdot = -2\alpha \frac{x}{k}$$

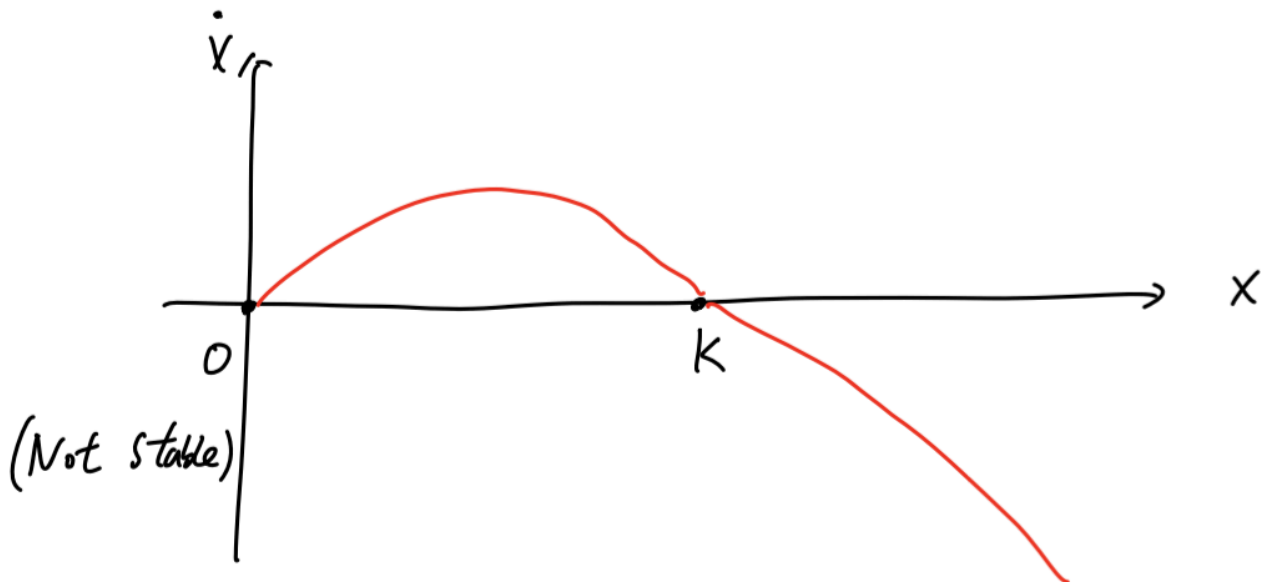
$$\dot{x} = \alpha(1 - \frac{x}{k})x = f(x) \text{ with } \alpha > 0, k > 0$$

$$f(x) = 0$$

$$\alpha(1 - \frac{x}{k})x = 0$$

$$x = 0 \text{ and } x = k$$

$x(0) = 0$ or k equilibrium points



Equilibrium points:

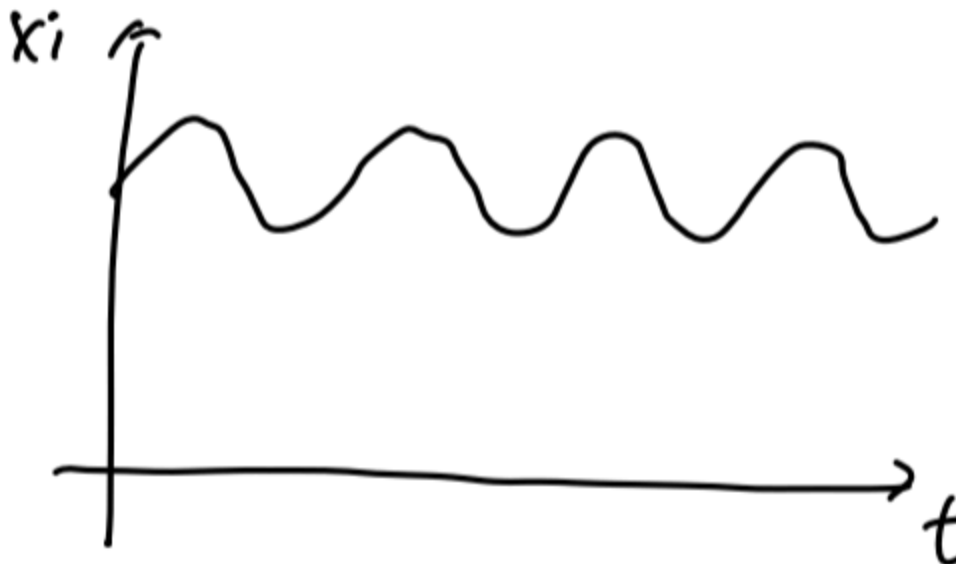
$$\dot{x} = f(x) \text{ --- } ()$$

Equilibrium points are constant trajectories

i.e., solution to $()$ that does not change with time



v.s.



$f(x) = 0 \implies x$ is an equilibrium point

$$f(x) = Ax$$

$$Ax = 0$$

-Always has equilibrium $x = 0$

-Could have infinitely many equilibriums, A is not invertible

$$A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$

$$\dot{x} = -x$$

$x = 0$ only equilibrium point

$$\dot{x} = -x + t$$

$$f(x, t) \implies x = t$$

$$\dot{x} = -tx$$

$x = 0$ is the only equilibrium point

Suppose $\dot{x} = f(x)$ has unique equilibrium of $x = k$

$$\dot{x} = x - k$$

$$\tilde{x} = x - k$$

$$\dot{\tilde{x}} = \dot{x}$$

$$\dot{\tilde{x}} = f(x)$$

$$\dot{\tilde{x}} = f(\tilde{x} + k)$$

Week 2

Linearization

$\dot{x} = f(x)$ time-invariant

x : a solution (feasible trajectory)

$$x(t) = x(t) + \tilde{x}(t)$$

where $x(t)$ is a given trajectory (e.g., an equilibrium)

$$\tilde{x}(t) = x(t) - x(t)$$

where $\tilde{x}(t)$ is the perturbation around x

Take derivative:

$$\dot{x}(t) = x + \dot{\tilde{x}}(t)$$

$$\dot{\tilde{x}}(t) = \dot{x}(t) - x(t)$$

$$\dot{\tilde{x}}(t) = f(x + \tilde{x}) - f(x) \text{ --- dynamics of the perturbed variable}$$

Because $\dot{x}(t)$ is a solution

$$\implies \dot{x}(t) = f(x(t))$$

Recall f is time invariant

- x is an equilibrium

$$\dot{\tilde{x}}(t) = f(x + \tilde{x}) - f(x)$$

where $f(x) = 0$

Time-invariant

- x is not an equilibrium

$$\dot{\tilde{x}} = f(x(t) + \tilde{x}) - f(x(t))$$

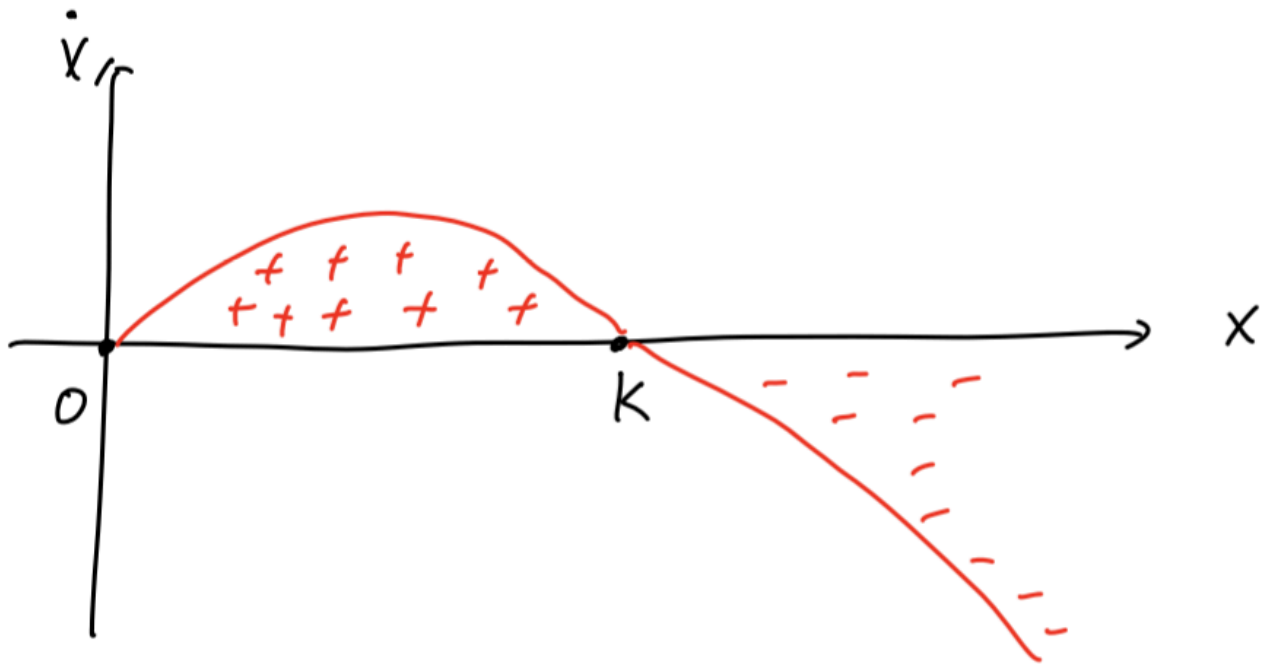
Time-varying

No approximation

$$\dot{\tilde{x}} = f(x + \tilde{x}) - f(x)$$

For sufficient small $\tilde{x}$

$$f(x + \tilde{x}) - f(x) \approx \frac{df}{dx} \bigg|_{x=x} \tilde{x} \text{ --- Taylor Approximation}$$



Approximate dynamics for the perturbed variable:

$$\dot{\tilde{x}} = \frac{f}{x} \bigg|_{x=x} \tilde{x}$$

where $\frac{f}{x} \bigg|_{x=x} = A$

$$\dot{\tilde{x}} = A\tilde{x}$$

$$\implies \tilde{x}(t) = e^{At}\tilde{x}(0) \text{ --- General solution}$$

$$e^{At} = 1 + At + \frac{A^2 t^2}{2} + \frac{A^3 t^3}{6} + \dots$$

Example. $\dot{x} = \alpha(1 - \frac{x}{k})x$

where $\alpha(1 - \frac{x}{k})x = f(x)$

$$\frac{f}{x} = \alpha(1 - \frac{x}{k})$$

$$\frac{f}{x} = \alpha(1 - \frac{2x}{k})$$

- $x = 0$

$$\frac{f}{x} \bigg|_{x=x=0} = \alpha$$

$$\dot{\tilde{x}} = \alpha\tilde{x}$$

$$\tilde{x}(t) = \tilde{x}(0)e^{\alpha t}$$

$$x = \tilde{x} + x = \tilde{x} + 0 = \tilde{x}$$

$$x(t) = x(0)e^{\alpha t}$$

- $x = k$

$$\frac{f}{x} \bigg|_{x=x=k} = -\alpha$$

$$\dot{\tilde{x}} = -\alpha\tilde{x}$$

$$\tilde{x}(t) = -\alpha\tilde{x}(0)e^{-\alpha t}$$

$$x = \tilde{x} + x$$

$$\tilde{x} = x - k$$

$$x(t) - k = (x(0) - k)e^{-\alpha t}$$

$$x(t) = k(1 - e^{-\alpha t}) + x(0)e^{-\alpha t}$$

$$x(t) = k + (x(0) - k)e^{-\alpha t}$$

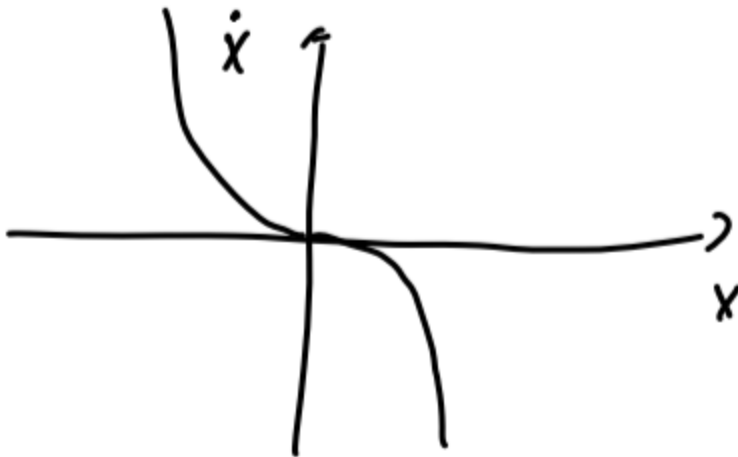
- Small perturbations around $x = 0$ are going to grow
 $x = 0$ is unstable
 - Small perturbations around $x = k$ decay with time
 $x = k$ is locally asymptotically stable
-

Ex.

1. $\dot{x} = -x$

$$f(x) = -x$$

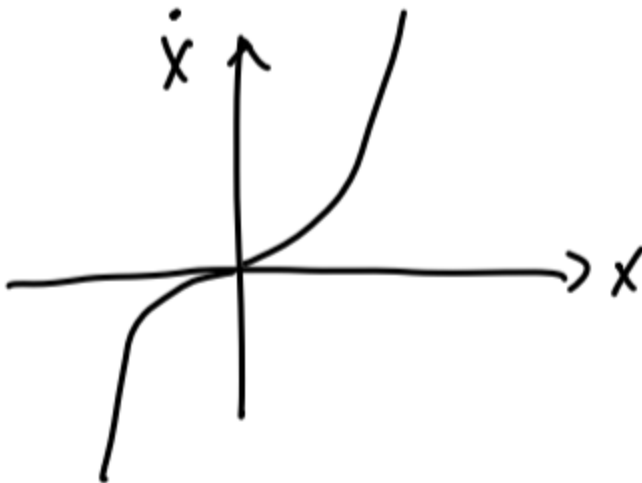
$$f(x) = 0 \implies x = 0$$



2. $\dot{x} = x$

$$f(x) = x$$

$$f(x) = 0 \implies x = 0$$



- $f = -x$
 $\frac{f}{x} = -x^2$
 $\frac{f}{x} \Big|_{x=0} = 0$

- $f = x$
 $\frac{f}{x} = x^2$
 $\frac{f}{x} x=0 = 0$

$$\dot{\tilde{x}} = 0 \quad \tilde{x} = 0 \quad \text{--- both systems}$$

$$\tilde{x}(t) = \tilde{x}(0)$$

$$x = \tilde{x} + x = \tilde{x} + 0 = \tilde{x}$$

$$x(t) = x(0)$$

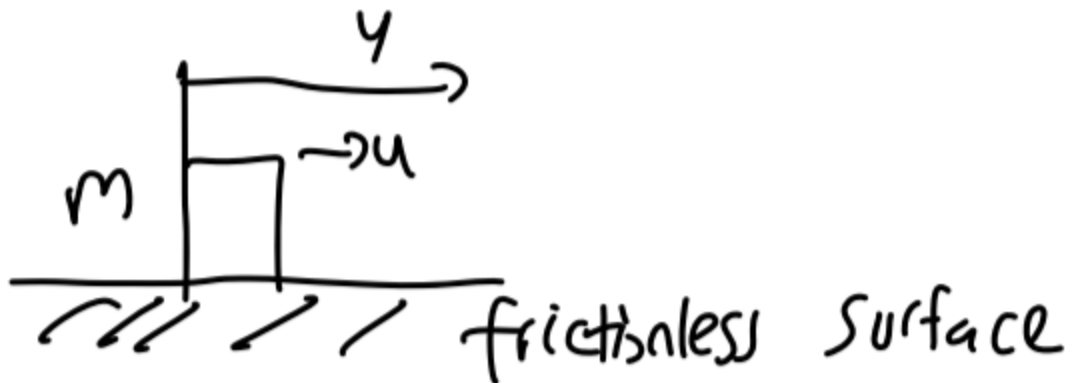
$$\dot{x}_1 = f_1(x) = f_1(x_1, x_2)$$

$$\dot{x}_2 = f_2(x) = f_2(x_1, x_2)$$

$$\frac{f}{x} = \frac{\frac{f_1}{x_1}}{\frac{f_2}{x_1}} \frac{\frac{f_1}{x_2}}{\frac{f_2}{x_2}}$$

$$x(t) = A^t x(0)$$

Ex1.



$$m =$$

$$= x_1$$

$$\dot{\cdot} = \dot{x}_1 = x_2$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \frac{1}{m}$$

$$\dot{x}_1 = 0 \quad 1 \quad x_1 + 0$$

$$\dot{x}_2 = 0 \quad 0 \quad x_2 + \frac{1}{m}$$

$$\text{where } A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \frac{1}{m}$$

$$\dot{x} = Ax +$$

$$\dot{x} = Ax +$$

$$\dot{x} = Ax +$$

$$\text{unforced system: } \dot{x} = Ax = f(x) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} x$$

$$x(t) = A^t x(0)$$

$$A^t = I + At + \frac{A^2 t^2}{2} + \dots$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} x_1 = 0$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} x_2$$

$$\implies x_2 = 0, x_1 \in \mathbf{R}$$

$$= 0 : x_1 = 0, x_2 = 0$$

$$x_1 = 1, x_2 = 0 \text{ (perturbation)}$$

$$A^2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A^2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A^t = I + At = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

$$A^t = I + At = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

$$x_1(t) = 1 - t x_1(0)$$

$$x_2(t) = 0 - 1 x_2(0)$$

$$x_2(t) = x_2(0)$$

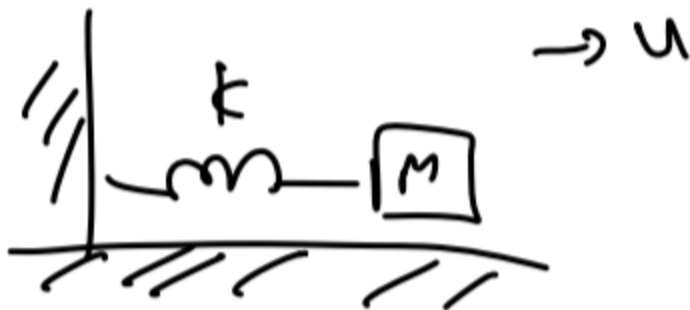
$$x_1(t) = x_1(0) + t x_2(0)$$

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$dt(A -) = dt \begin{pmatrix} - & 1 \\ 0 & - \end{pmatrix} = 2 = 0$$

$$\Rightarrow = 0$$

Ex2.



$$m + k =$$

$$A = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{pmatrix}, \quad = \frac{1}{m}$$

$$dt(A -) = dt \begin{pmatrix} - & 1 \\ -\frac{k}{m} & - \end{pmatrix} = 2 + \frac{k}{m} = 0$$

$$\Rightarrow = \frac{k}{m}$$

Ex3.



$$m + \cdot + k =$$

$$= x_1$$

$$x_2 = \dot{x}_1$$

$$\dot{x}_2 = \frac{1}{m} (-kx_1 - x_2)$$

$$\dot{x}_1 = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{1}{m} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \frac{1}{m}$$

$$A = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{1}{m} \end{pmatrix}, \quad = \frac{1}{m}$$

$$A = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{1}{m} \end{pmatrix}, \quad = \frac{1}{m}$$

$$dt(A -) = dt \left(-\frac{k}{m} \quad -\frac{1}{m} \right) = \lambda^2 + \frac{k}{m} = 0$$

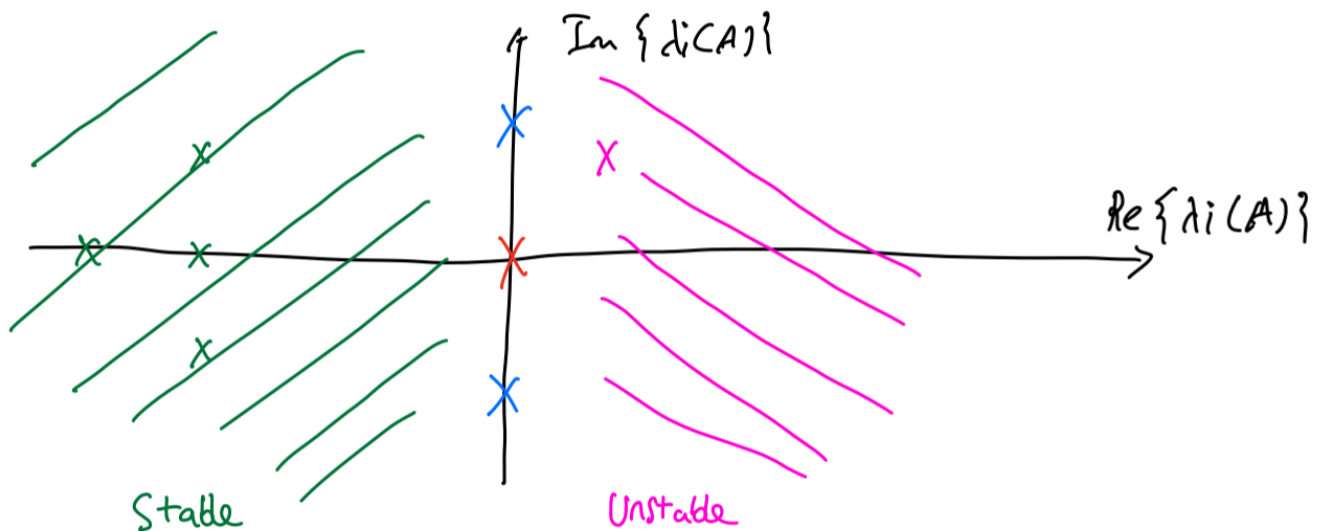
$$\Rightarrow \lambda = \frac{-\frac{k}{m} \pm \sqrt{\frac{k^2}{m^2} - \frac{4}{m}}}{2}$$

$$= \frac{-\frac{k}{m} \pm \sqrt{\frac{k^2 - 4m}{m^2}}}{2}$$

$$= \frac{-k \pm \sqrt{k^2 - 4m}}{2m}$$

$$\lambda_{1,2} = 0$$

$$(1,2) \quad 0$$



All e-values strictly in the left half plane $\Rightarrow x = 0$ is stable

("every trajectory converges to $x = 0$ ")

At least one-value in the right half plane $\Rightarrow x = 0$ is not stable

("some trajectories grow unbounded")

2D systems

characteristic polynomial: $\lambda^2 + 1 + 0 = 0$

$\lambda_{1,2} = \pm i \Rightarrow$ stability

Range of phenomena for $\dot{x} = f(x)$ (nonlinear)

1. Multiple "isolated" equilibrium points
2. Finite escape time
3. Limit cycles
4. Chaos

Ex1.



not an isolated equilibrium

- If a neighborhood around x which does not contain any other equilibrium
 $\implies x$ is an isolated equilibrium
- All neighborhoods around x contain at least one other equilibrium
 $\implies x$ is not an isolated equilibrium

Class3

$$\dot{x} = f(x)$$

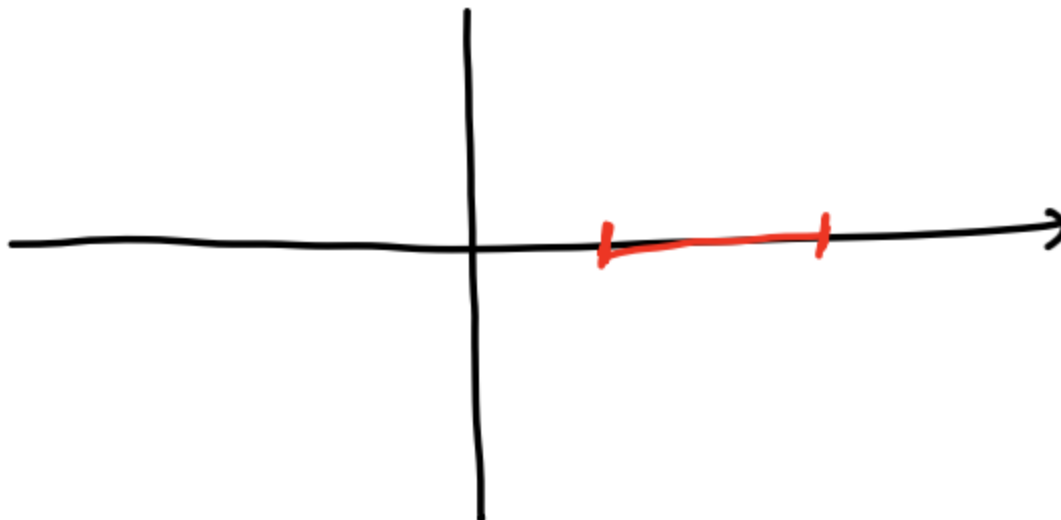
Definition: $x(t)$ is a periodic orbit if $T > 0$ such that $x(t+T) = x(t), t$

Linear systems:

1-Dimensional:

$$\dot{x} = x$$

No periodic orbits



$f(x)$ along $x(t)$ is monotonic

2-Dimensional:

$$\dot{x} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} x$$

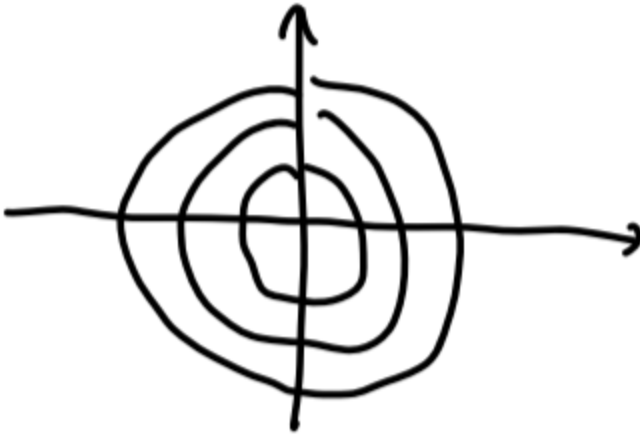
$$1,2 =$$

$$\dot{x}_1 = -x_2$$

$$\dot{x}_2 = x_1$$

$$x_1 = -\dot{x}_2$$

$$x_2 = \dot{x}_1$$



$$\begin{aligned} \frac{d}{dt}(x_{21} + x_{22}) &= 2x_1\dot{x}_1 + 2x_2\dot{x}_2 \\ &= 2x_1(-x_2) + 2x_2(x_1) \\ &= 0 \end{aligned}$$

an "isolated" periodic orbit cannot occur in linear system

limit cycle = "isolated" periodic orbit

$$\dot{x} = Ax$$

2-Dimensional:

$$= T^{-1}x$$

$$\implies \dot{x} = T^{-1}\dot{x} = T^{-1}Ax = T^{-1}AT$$

For proper choice of T

$$\dot{x} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} x$$

or

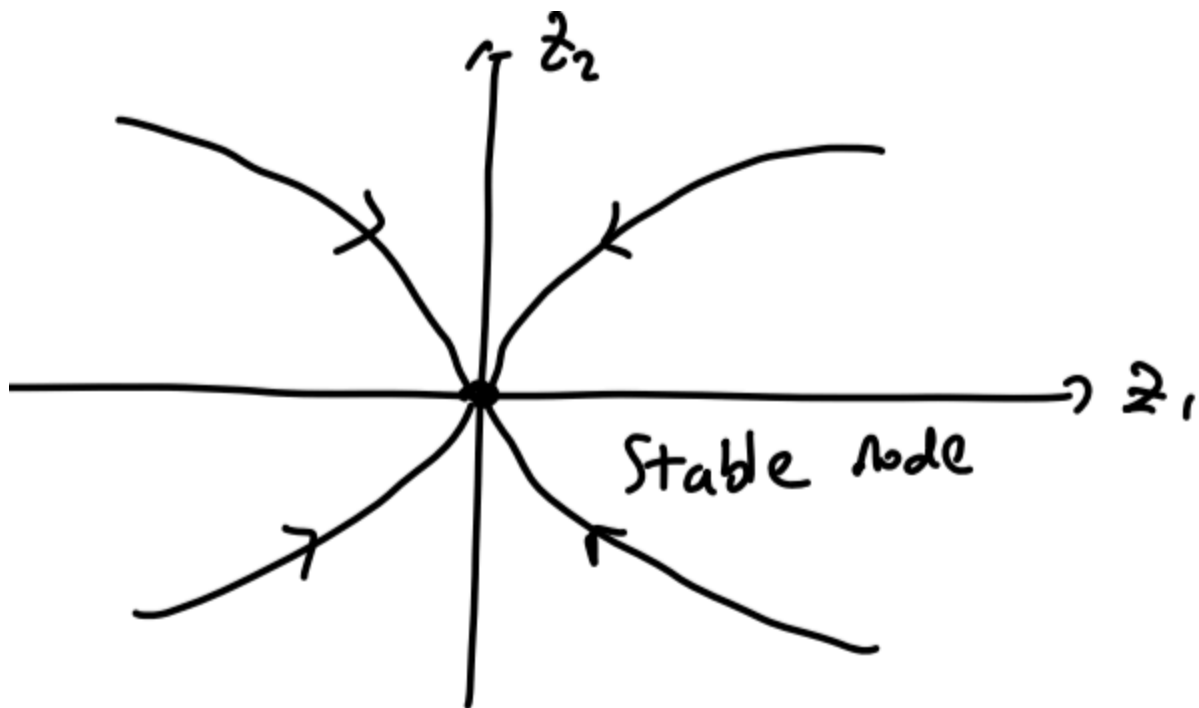
$$\dot{x} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} x$$

or

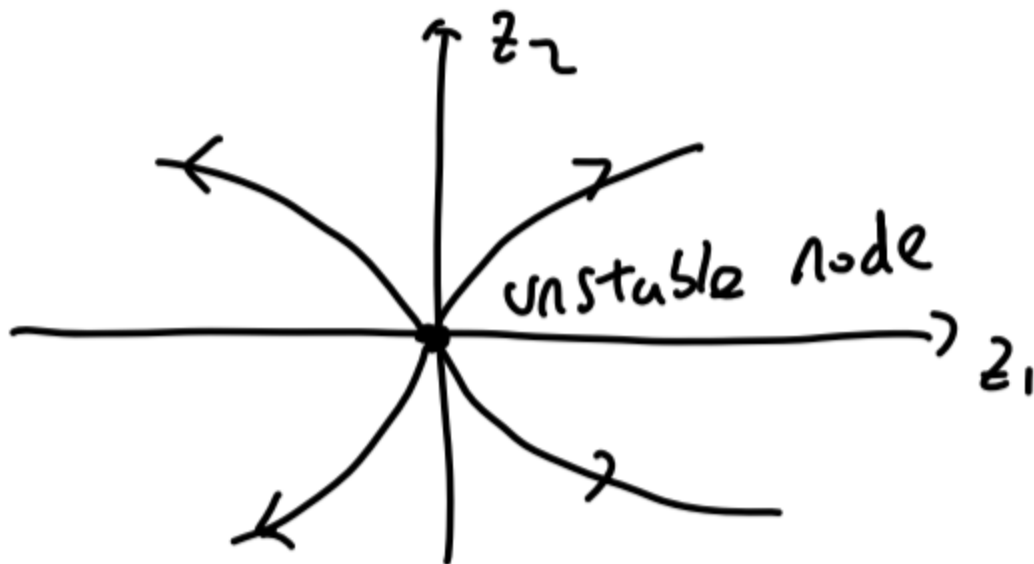
$$\dot{x} = \begin{pmatrix} \alpha & \\ - & \alpha \end{pmatrix} x$$

where $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, $\begin{pmatrix} \alpha & \\ - & \alpha \end{pmatrix}$ are $T^{-1}AT$

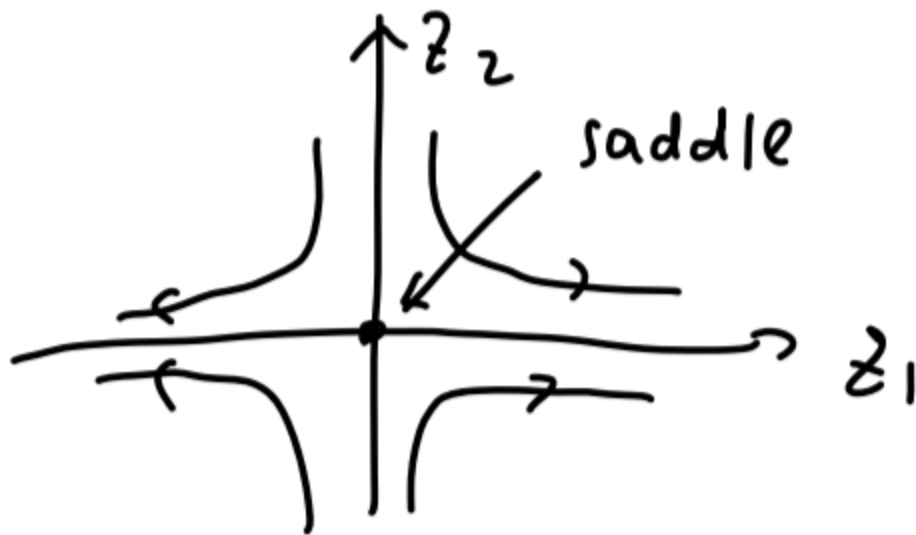
Real



$$\begin{aligned} \dot{\mathbf{z}} &= \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \mathbf{z} \\ \mathbf{z}_1 &= \mathbf{z}_1(0) \\ \mathbf{z}_2 &= \mathbf{z}_2(0) \end{aligned}$$



$$\begin{aligned} \dot{\mathbf{z}} &= \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \mathbf{z} \\ \mathbf{z}_1 &= \mathbf{z}_1(0) \\ \mathbf{z}_2 &= \mathbf{z}_2(0) \end{aligned}$$



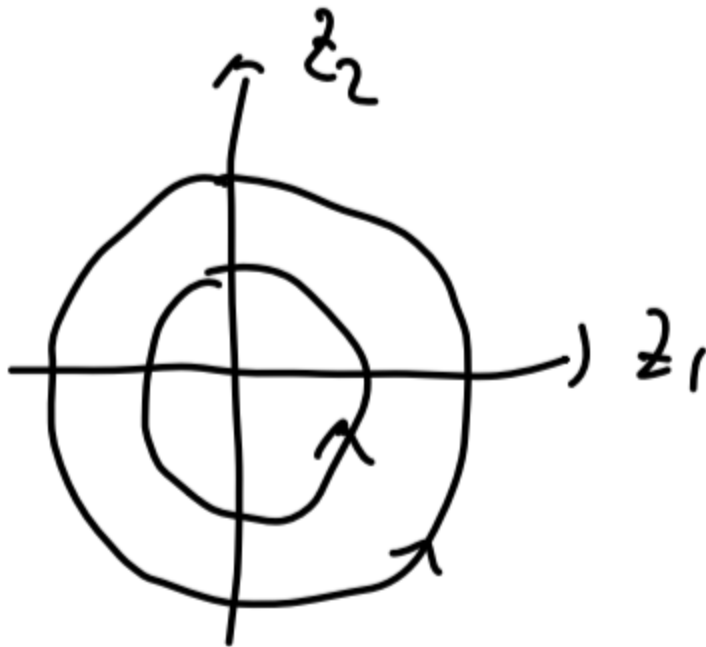
$$\begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$$

Complex eigenvalues

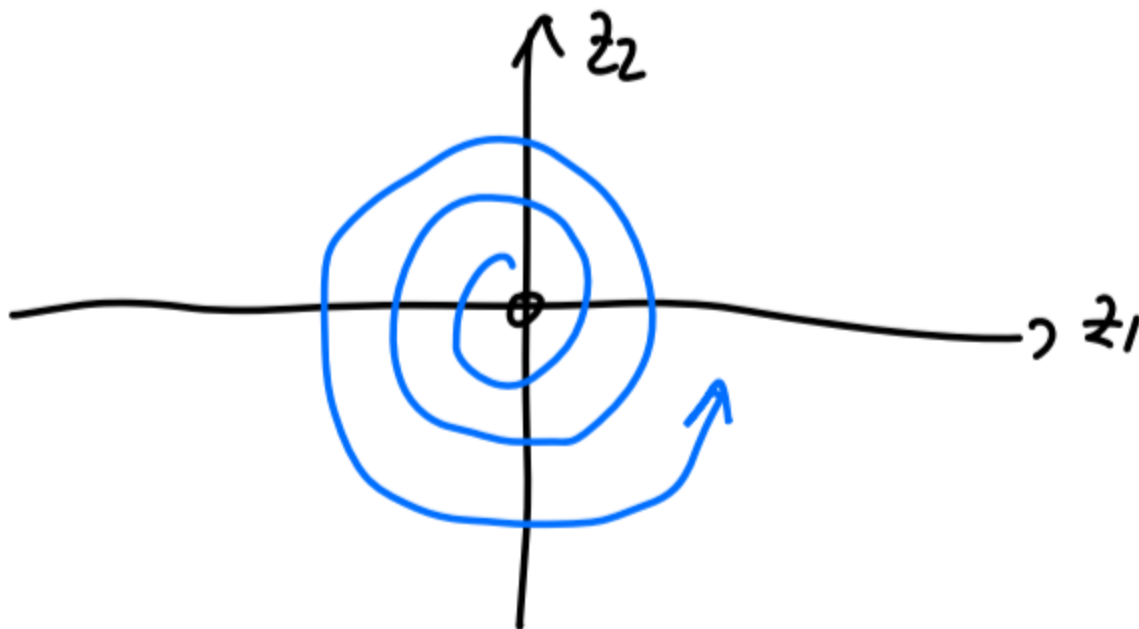
$$\dot{z} = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix} z$$

$$\alpha = 0$$

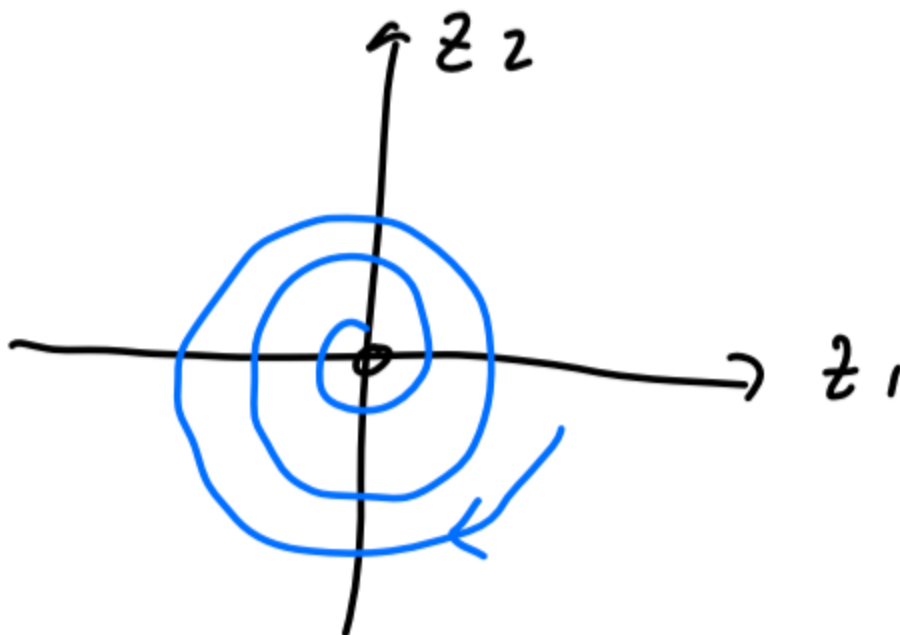
"marginally stable"



$$\begin{aligned} \frac{d}{dt} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} &= \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \\ &= \begin{pmatrix} \alpha z_1 + \beta z_2 \\ -\beta z_1 + \alpha z_2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \end{aligned}$$



$\alpha > 0$



$$\dot{x} = Ax, A \in \mathbf{R}^{n \times n}$$

(i) $\operatorname{Re}(\lambda_i) < 0 \implies$ stable equilibrium

(ii) $\operatorname{Re}(\lambda_i) > 0$ for at least one $i \implies$ unstable equilibrium

(iii) $\operatorname{Re}(\lambda_i) = 0$ and no repeated eigenvalues on the imaginary axis $\implies$ "marginally stable"

Ex. $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$\lambda_{1,2} = 0$ "unstable system"

$$\begin{aligned} \eta_1 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \eta_2 &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} x_1(t) &= x_1(0)t + x_1(0) \\ x_2(t) &= x_2(0) \end{aligned}$$

Hartman-Grobman Theorem

Provides basis for qualitative comparison between $\dot{x} = f(x)$ and $\dot{x} = \frac{f}{x}(x-x)$

Theorem: If x is a hyperbolic (i.e., no eigenvalue on the imaginary axis) equilibrium

Then there exists a **homeomorphism**

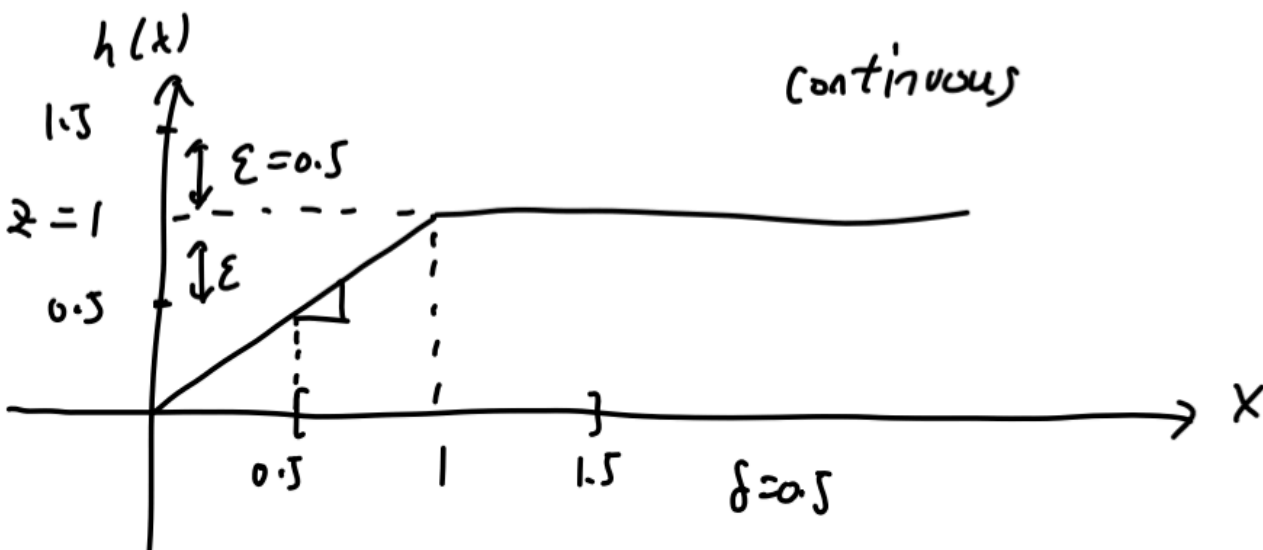
$h(x)$ defined in a neighborhood of x that maps trajectories of $\dot{x} = f(x)$ to those of

$$\dot{y} = A, A = \frac{f}{x}(x-x)$$

Homeomorphism: a continuous map with a continuous inverse

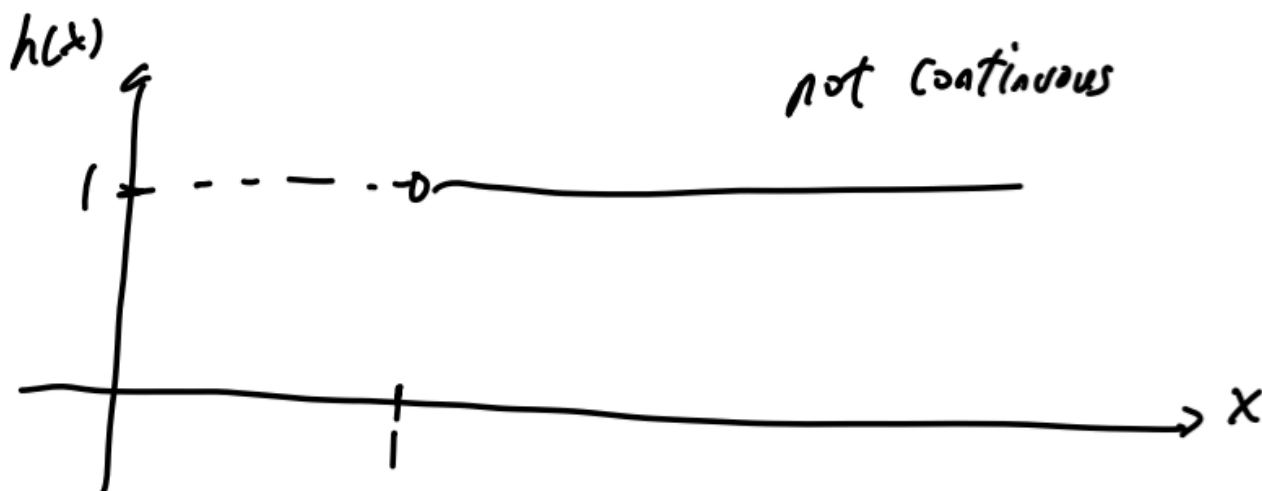
$h(x)$ is continuous at x if $\forall \epsilon > 0$ such that $\|h(x) - h(x_0)\| < \epsilon$

$h(x_0) = h(x_0)$



$$\|h(x) - h(x_0)\| < \epsilon$$

where $x_0 = 1, \|x - x_0\| = 1, \|x - x_0\| = 0, \|x - x_0\| = 0$.

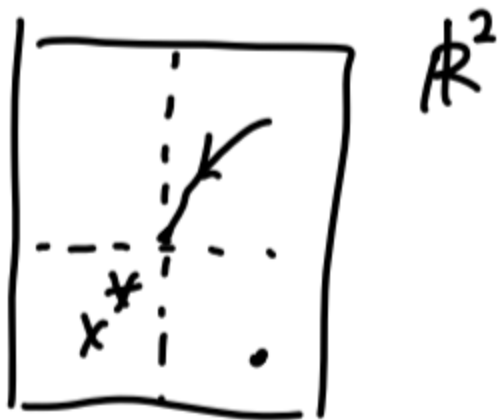


$\epsilon = 0$. (user specified)

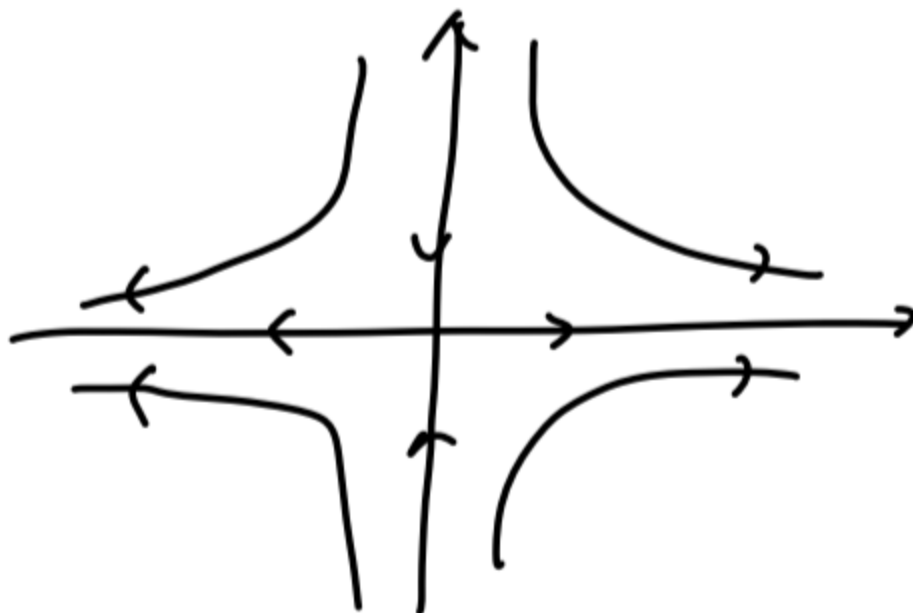
Pick

$$\|h(x) - h(x_0)\| < \epsilon$$

$\cdot = A$

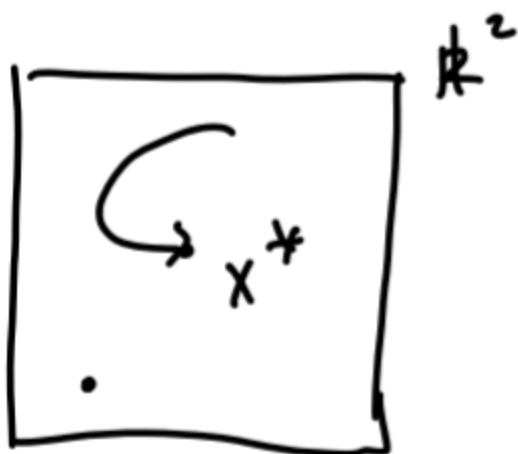


$= (x)$



saddle point

$\dot{x} = f(x)$





Ex. necessity of hyperbolicity condition

$$\dot{x}_1 = -x_2 + x_1(x_{21} + x_{22})$$

$$\dot{x}_2 = x_1 + x_2(x_{21} + x_{22})$$

$$x_1 = x_2 = 0$$

$$\frac{f_1}{x_1} = x_{21}x_{22} = 0$$

$$\frac{f_1}{x_2} = 2x_1x_2 - 1_{x=x} = -1$$

$$\frac{f_2}{x_1} = 2x_1x_2 + 1_{x=x} = 1$$

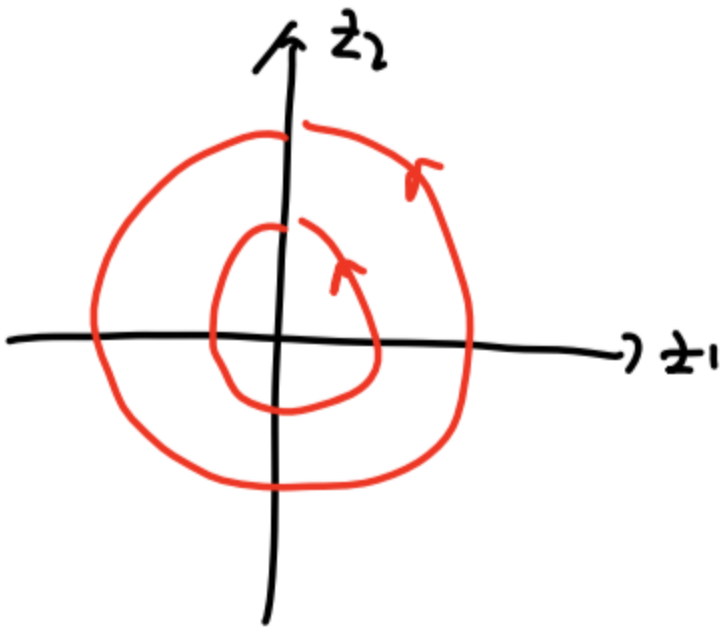
$$\frac{f_2}{x_2} = x_{22}x_{22} = 0$$

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$^2 + 1 = 0$$

$$=$$

$$\cdot = Ax$$



$(nt) - ()$ is very small

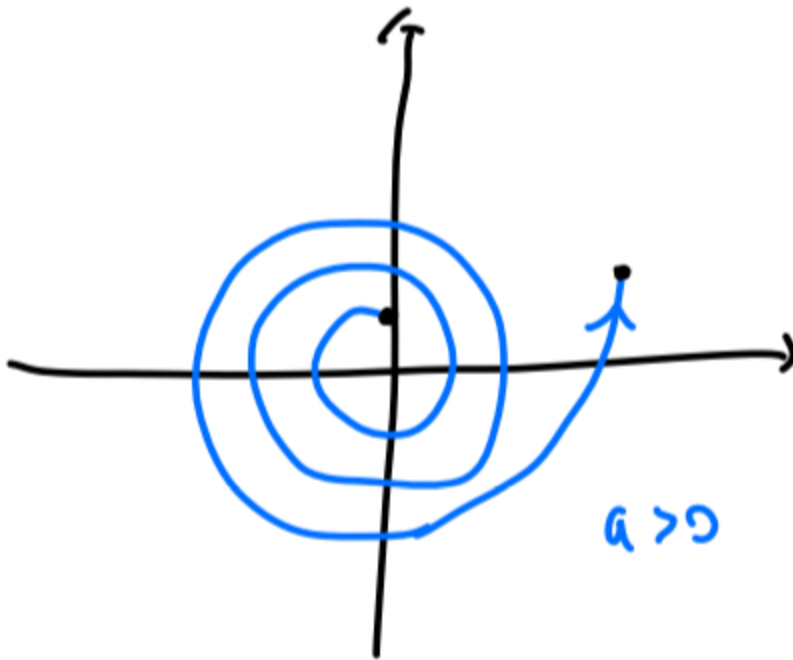
$$x(nt) - x()$$

$$\dot{x} = f(x)$$

$$= x_{21} + x_{22}$$

$$= -1\left(\frac{x_2}{x_1}\right)$$

$$\dot{x} = 1$$



$$\dot{x} = f(x)$$

- $\dot{x} = 0$ is locally asymptotically stable
- $\dot{x} = 0$ is unstable
 x_0 such that $x(t)$ grows unbounded

$$\dot{x} = A$$

$$A = \frac{f}{x} \bigg|_{x=\dot{x}=0}$$

- $(i) \ 0, i$, locally asymptotically stable
- $(i) \ 0$ for some i , unstable

This is the end of linearization

1 Dimensional - Finite escape time

$\dot{x} = x^2$, $x = 0$ is the equilibrium point

$$x x_0 \frac{dx}{x^2} = \int_{t_0}^t dt$$

$$-\frac{1}{x} \bigg|_{x_0}^x = t - t_0$$

$$\frac{1}{x_0} - \frac{1}{x} = t - t_0$$

$$\implies \frac{1}{x} = \frac{1}{x_0} - (t - t_0)$$

$$\frac{1}{x} = \frac{1 - x_0(t - t_0)}{x_0}$$

$$x = \frac{x_0}{1 - x_0(t - t_0)}$$

$$t_0 = 0, x(t) = \frac{1}{\frac{1}{x_0} - t}$$

$$t \rightarrow \frac{1}{x_0}$$

$$\implies x(t) \rightarrow$$

Ex.

$$\dot{x} = x^2 + 1, x \in \mathbf{R}$$

No equilibrium

Class 4

Periodic orbits in the plane

Bendixson's Theorem:

Time-invariant planar system:

$$\dot{x}_1 = f_1(x_1, x_2)$$

$$\dot{x}_2 = f_2(x_1, x_2)$$

$$\nabla f(x) = \frac{f_1}{x_1} + \frac{f_2}{x_2}$$

If $\nabla f(x)$ is not identically zero and does not change sign in a simply connected region then there are no periodic orbits exist that lie entirely in



simply connected but not convex

For a linear system

$$f(x_1, x_2) = A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$A = \begin{pmatrix} \frac{f_1}{x_1} & \frac{f_1}{x_2} \\ \frac{f_2}{x_1} & \frac{f_2}{x_2} \end{pmatrix}$$

$$\nabla f(x) = \frac{f_1}{x_1} + \frac{f_2}{x_2} = T(A)$$

For a periodic orbit to exist in a linear system, it is necessary for $T(A) = 0$

Periodic orbit:

$$\dot{x} = \begin{pmatrix} 0 & - \\ & 0 \end{pmatrix} x$$

$$\dot{x} = \begin{pmatrix} \alpha & - \\ & \alpha \end{pmatrix} x, \alpha \neq 0$$

$$T(A) = 2\alpha$$

$$\dot{x} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} x$$

$$T(A) = 0$$

Ex.

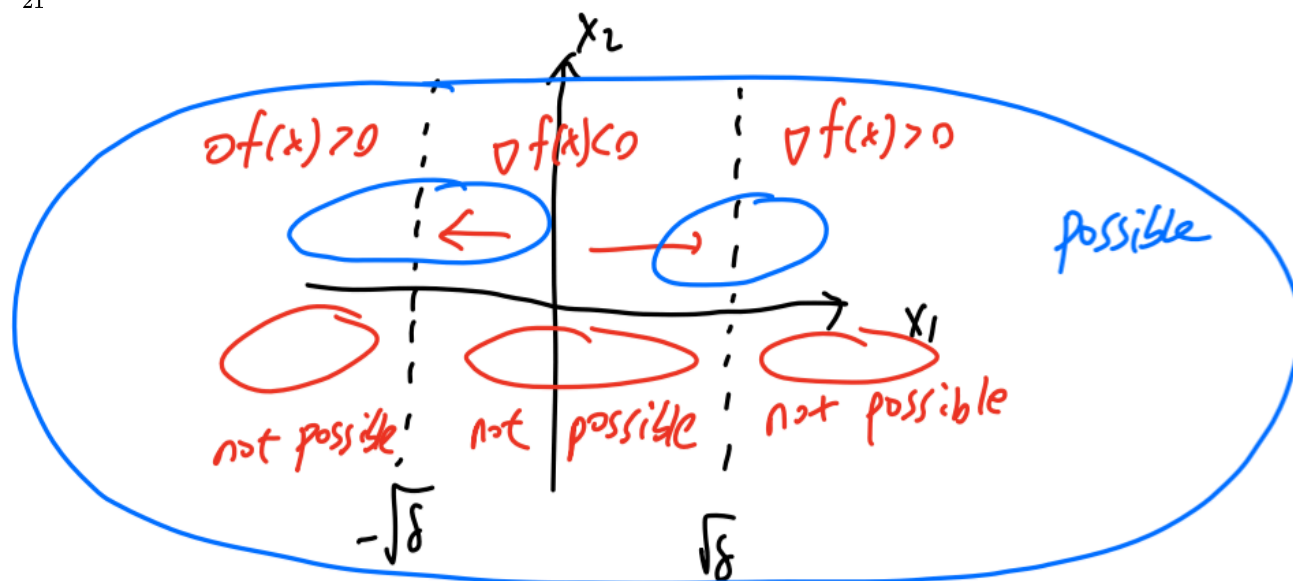
$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_2 + x_1 - x_1 + x_2 x_2$$

$$\nabla f(x) = \frac{f_1}{x_1} + \frac{f_2}{x_2}$$

$$= 0 + (- + x_{21})$$

$$= x_{21} -$$

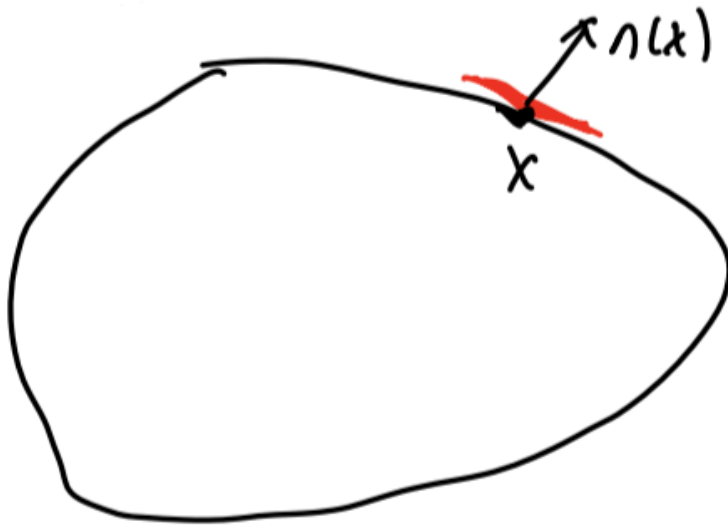


Poincare-Bendixson Theorem

Invariant set

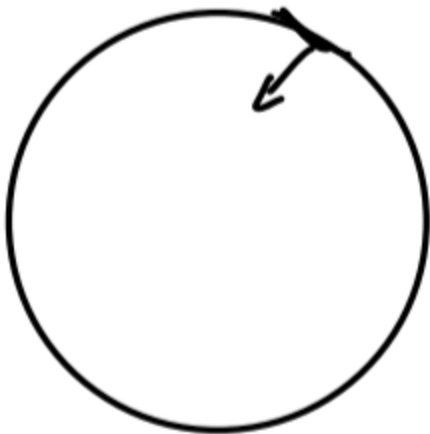
A set $M \subset \mathbb{R}^n$ is (positively) invariant, if for each initial condition in M , the trajectory remains in M , $t \geq 0$





$$\dot{x} = f(x)$$

$f(x) \cdot n(x) \leq 0, x$ on the boundary of $\Rightarrow$ is positively invariant

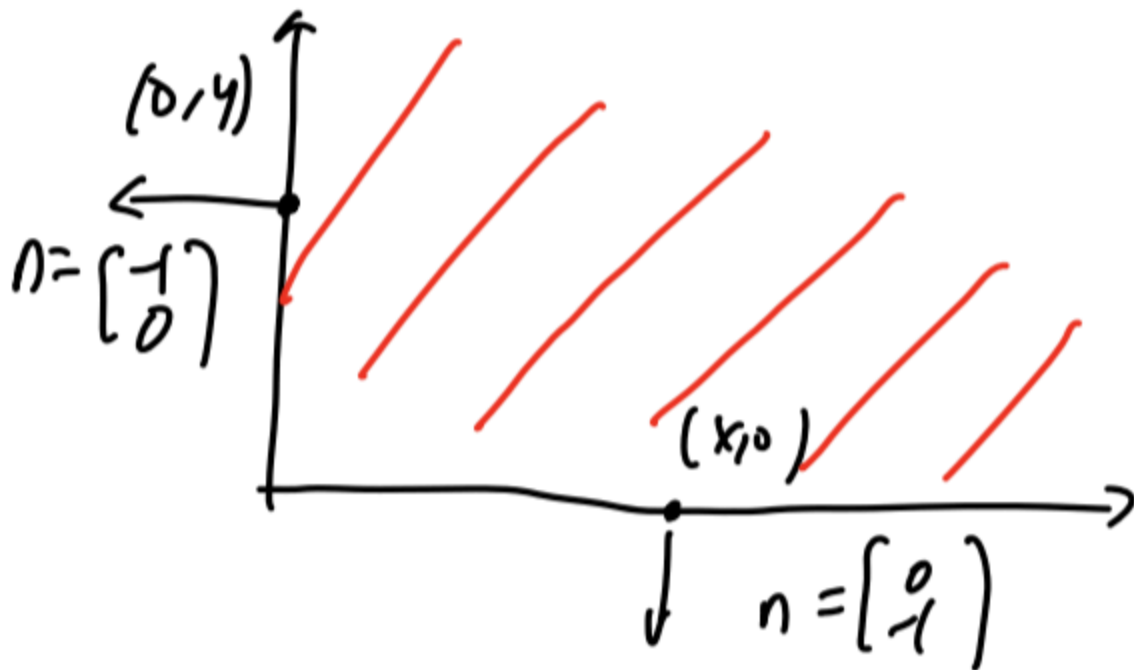


$f(x) \cdot n(x) \leq 0, x$ on the boundary of this circle $\Rightarrow$ circle is positively invariant
under-approximation of the maximal positively invariant set

Ex. Predator-Prey model, $a, b, c, d \geq 0$

$\dot{x} = (a - b)x$ --- Prey (exponential growth when $y = 0$)

$\dot{y} = (c - d)y$



$$f \cdot n = 0$$

$$(x-d)(-1) = 0 \text{ for } x = 0$$

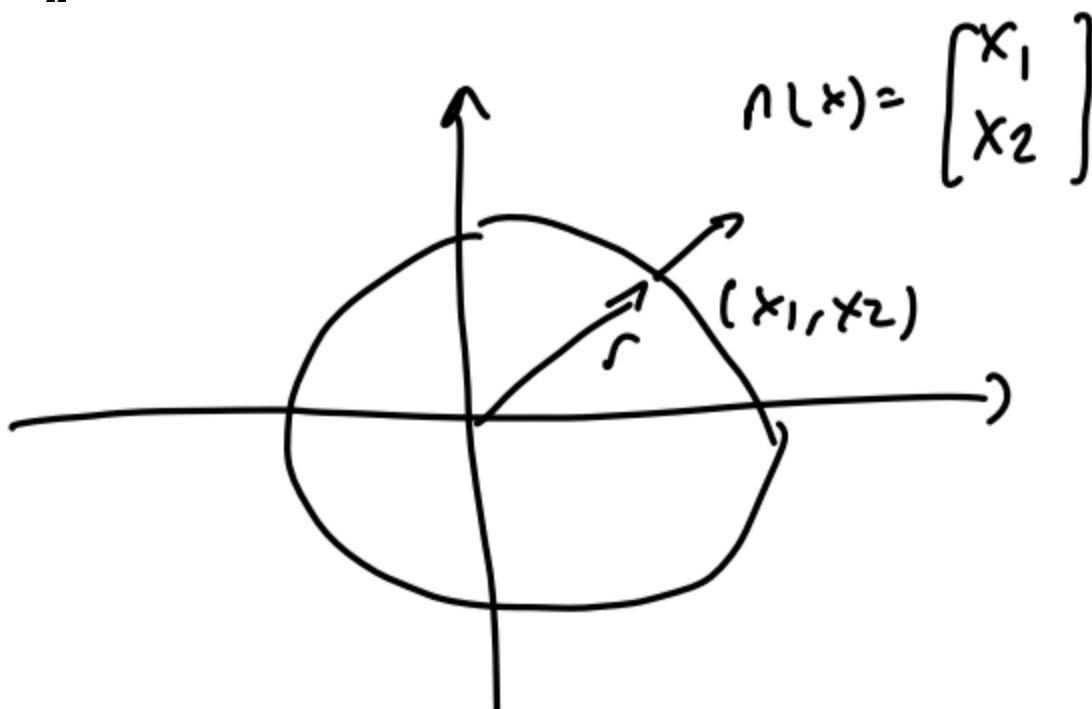
$$= 0$$

Ex.

$$\dot{x}_1 = x_1 + x_2 - x_1(x_{21} + x_{22})$$

$$\dot{x}_2 = -2x_1 + x_2 - x_2(x_{21} + x_{22})$$

$x : x_{21} + x_{22}$ is positively invariant for sufficiently large



$$f(x) \cdot n(x) = x_{21} + x_1x_2 - x_{21}(x_{21} + x_{22}) - 2x_1x_2 + x_{22} - x_{22}(x_{21} + x_{22})$$

$$= x_{21} + x_{22} - (x_{21} + x_{22})^2 - x_1x_2$$

$$= \frac{1}{2} - \frac{1}{2}x_1x_2$$

$$(x_1 + x_2)^2 = 0$$

$$\begin{aligned}
 & x_{21} + x_{22} + 2x_1x_2 = 0 \\
 & -x_1x_2 = \frac{x_{21}+x_{22}}{2} \\
 & -x_1x_2 = \frac{1}{2} \\
 & f(x) = \frac{1}{2} - x_1x_2 \\
 & = \frac{1}{2} - \frac{1}{2} \\
 & = 0 \\
 & 0 \text{ if } \frac{1}{2}
 \end{aligned}$$

Poincare-Bendixson Theorem

Suppose S is compact (closed and bounded), and positively invariant for a planar time invariant system $\dot{x} = f(x), x \in \mathbb{R}^2$

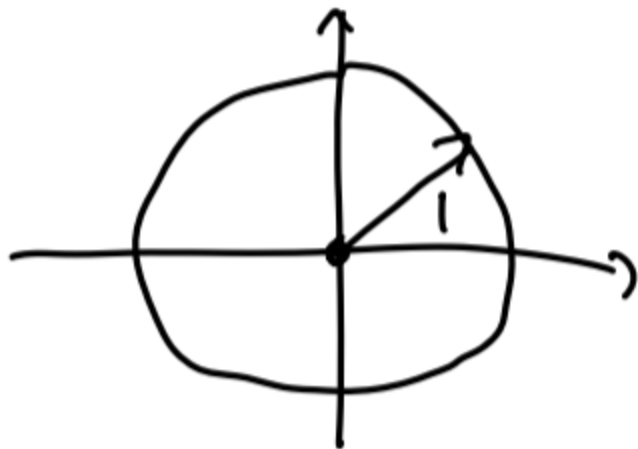
If S contains no equilibrium points, then it contains a periodic orbit

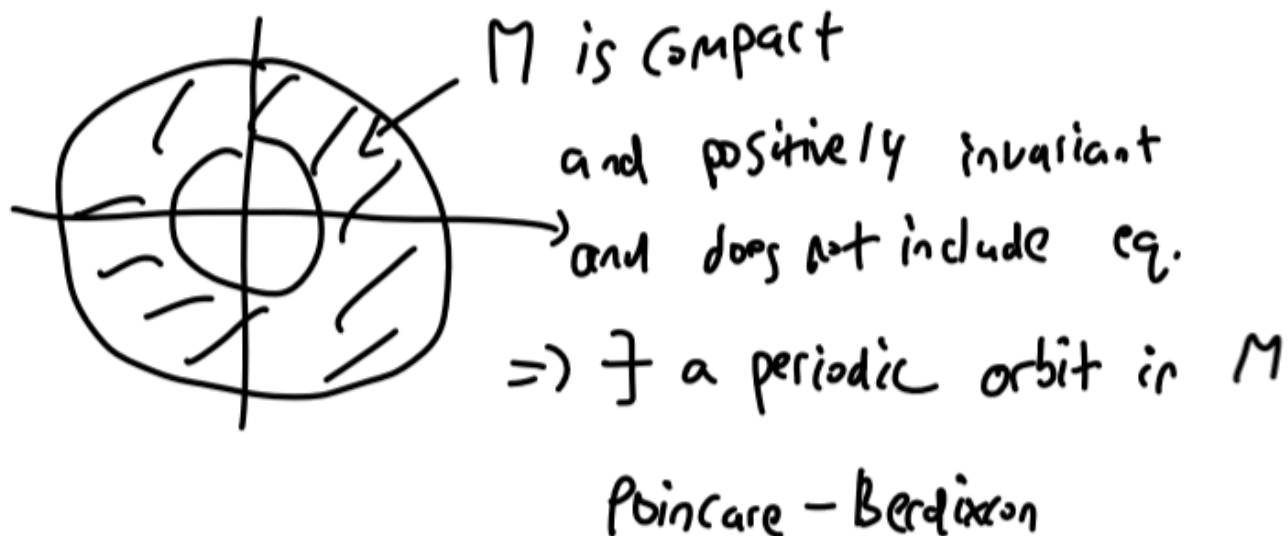
Ex.

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\dot{x}_1 = -x_2$$

$$\dot{x}_2 = x_1$$





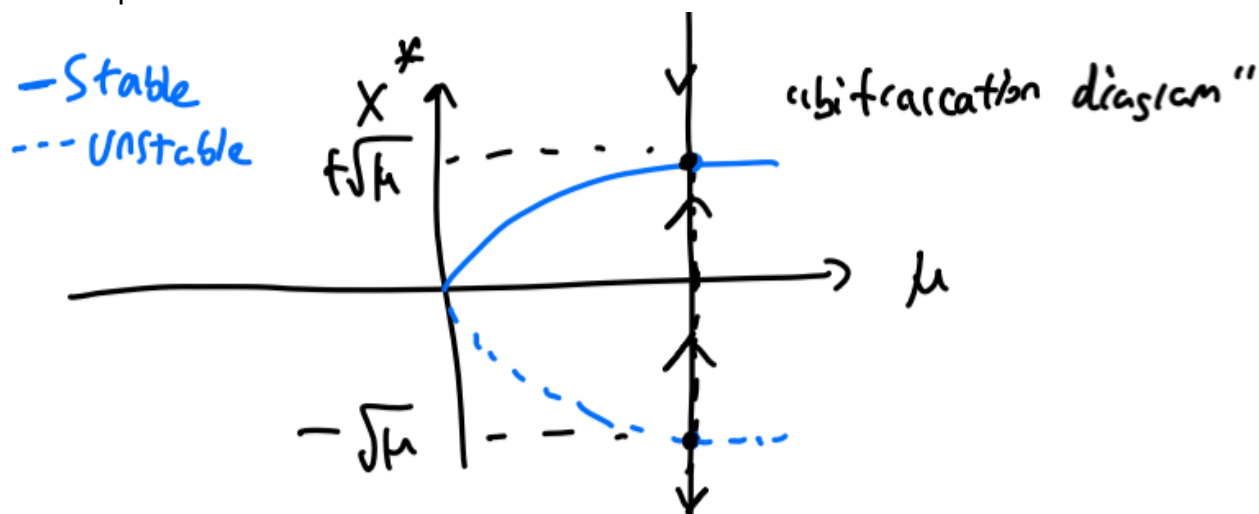
is compact and positively invariant and does not include equilibrium
 $\Rightarrow$ a periodic orbit in

Bifurcations

A bifurcation is an abrupt change in qualitative behavior as a parameter is varied.

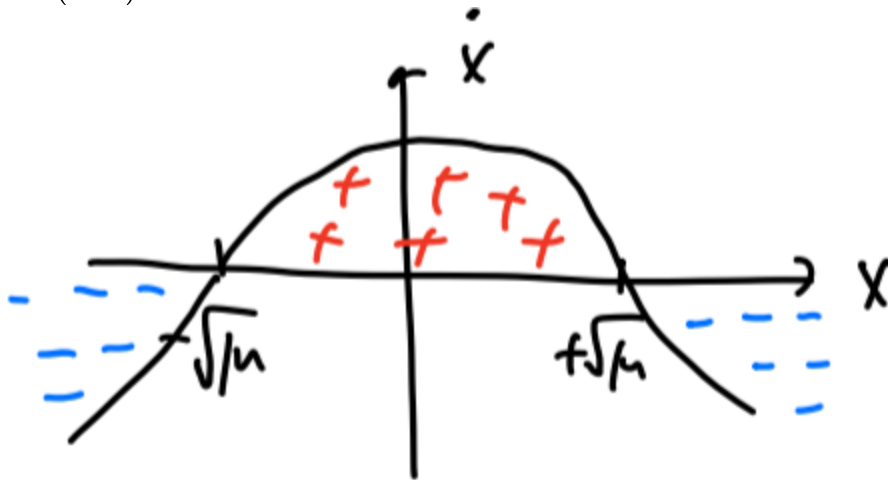
Example: equilibria or limit cycles appearing/ disappearing becoming stable/ unstable

- Transition: multiple equilibria no equilibrium
- Fold bifurcation, "blue-sky" bifurcation, or "saddle node" bifurcation
- $$\dot{x} = -x^2 = f(x)$$
- $f(x) = 0$ has solution only if 0
- $0 < \mu$: no equilibrium
 $\mu = 0$: one equilibrium
 $0 > \mu$: two equilibria



$$\begin{aligned}\dot{x} &= -x^2 \\ \dot{\tilde{x}} &= \frac{f}{x} \bigg|_{x=\tilde{x}} (\tilde{x} - x) \\ \dot{x} &= \end{aligned}$$

$$\begin{aligned}\frac{f}{x} &= \frac{1}{x}(-x^2) \\ &= -x \\ &= -\sqrt{\mu} \\ \dot{\tilde{x}} &= -2(\tilde{x} - \sqrt{\mu})\end{aligned}$$



- Trans-critical bifurcation

$$\dot{x} = x - x^2 = f(x)$$

$$x - x^2 = 0$$

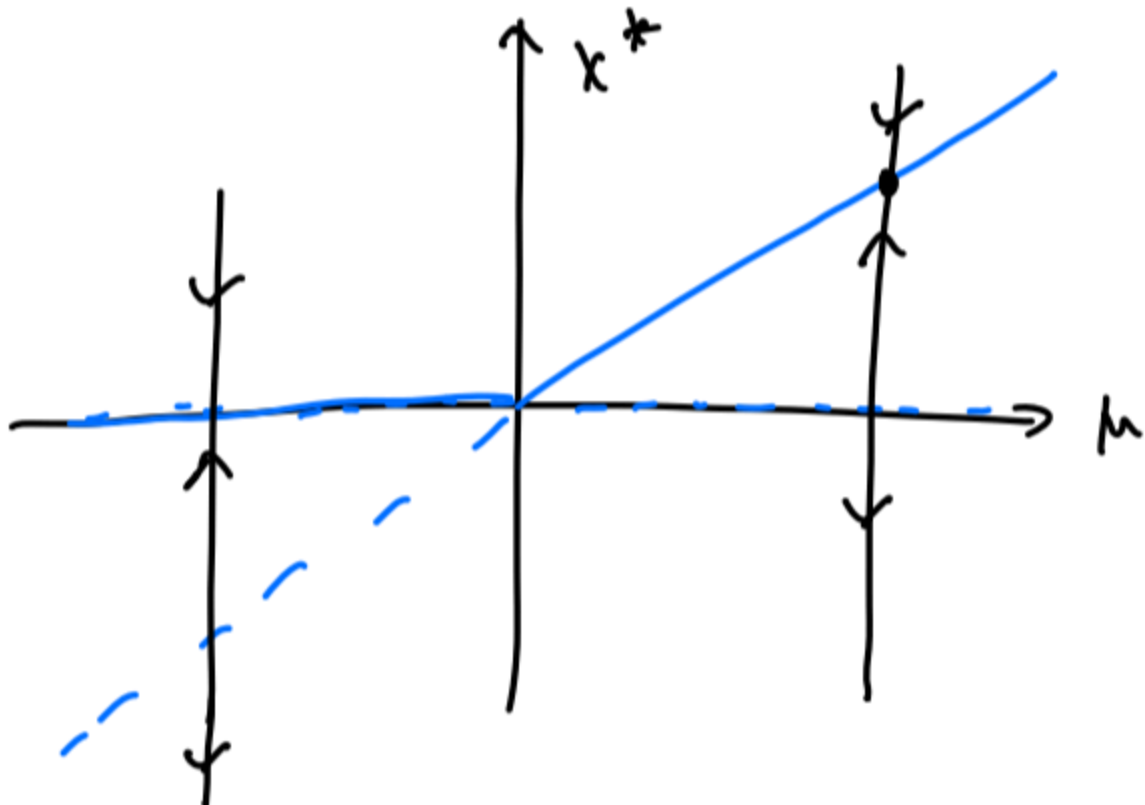
$$\implies x = 0 \text{ and } x =$$

0 : $x = 0$ is stable, $x =$ is stable

0 : $x = 0$ is unstable, $x =$ is stable

$$\frac{f}{x} \bigg|_{x=0} = -2x \bigg|_{x=0} =$$

$$\frac{f}{x} \bigg|_{x=} = -2x \bigg|_{x=} = -$$



- Pitchfork Bifurcation

$$\dot{x} = x - x$$

$$x - x = 0$$

$0 : x = 0$ always

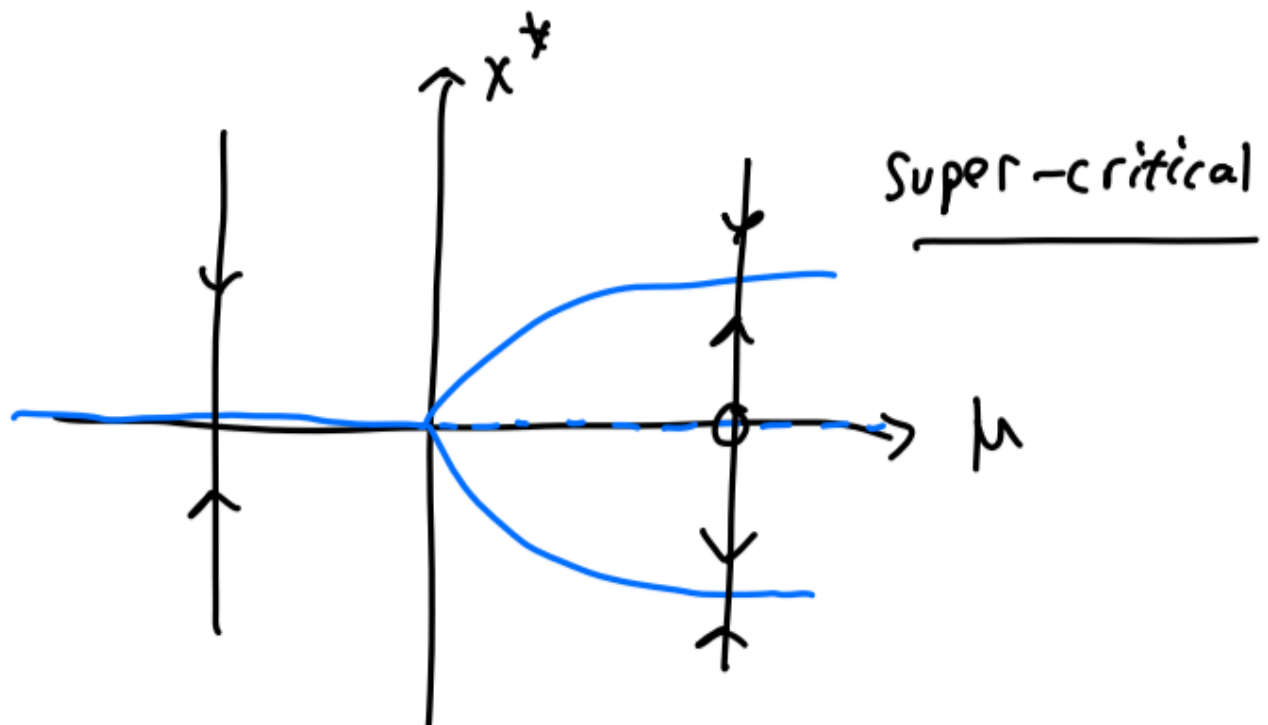
$0 : x = 0, x = x$

	0	0
$x = 0$	Stable	Unstable
$x = +$	N/A	Stable
$x = -$	N/A	Stable

$$\frac{f}{x} x=0 = -x^2_{x=0} =$$

$$\frac{f}{x} x=+ = -x^2_{x=+} = -2$$

$$\frac{f}{x} x=- = -x^2_{x=-} = -2$$



super-critical

$$\dot{x} = x + x$$

